

# Dipal Achalesh Thaker

Illinois, USA | [dipalat2@illinois.edu](mailto:dipalat2@illinois.edu) | [linkedin.com/in/dipalthaker2](https://www.linkedin.com/in/dipalthaker2) | [github.com/dipalthaker](https://github.com/dipalthaker) | [dipalthaker.github.io/dipal-portfolio](https://dipalthaker.github.io/dipal-portfolio)

## TECHNICAL SKILLS

**Languages:** Python, Java, C++, JavaScript, TypeScript, SQL, JSON  
**AI, Automation & Data:** LLM APIs, RAG Pipelines, LlamaIndex, Workflow Automation, Data Validation, Pandas, NumPy  
**Backend & APIs:** FastAPI, Node.js, Spring Boot, REST APIs, GraphQL, Microservices, Pydantic, Swagger  
**Databases:** PostgreSQL, MySQL, SQLite, SQL ETL, Query Optimization, Data Modeling  
**Cloud & DevOps:** Docker, Kubernetes, AWS EC2, AWS S3, AWS Lambda, GitHub Actions, Jenkins, CI/CD  
**Frontend & Web:** React, Angular, Flutter, HTML, CSS, JavaScript, TypeScript, Redux  
**Testing & Monitoring:** Pytest, Jest, Prometheus, Grafana, Root-Cause Analysis

## EDUCATION

|                                                                                                               |                                                         |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>University of Illinois Urbana Champaign</b><br><i>MS in Information Management (STEM)   GPA: 3.89/4.00</i> | <i>Aug 2024 – May 2026</i><br><i>Illinois, USA</i>      |
| <b>K. J. Somaiya College of Engineering</b><br><i>B.Tech in Information Technology   GPA: 3.40/4.00</i>       | <i>Aug 2019 – Jul 2023</i><br><i>Maharashtra, India</i> |

## PROFESSIONAL EXPERIENCE

|                                                                                         |                                                    |
|-----------------------------------------------------------------------------------------|----------------------------------------------------|
| <b>Software Engineer AI &amp; Full Stack</b><br><i>Business Intelligence Group UIUC</i> | <i>Aug 2025 – Dec 2025</i><br><i>Illinois, USA</i> |
|-----------------------------------------------------------------------------------------|----------------------------------------------------|

- Built 2 LLM-powered RAG chatbots for 40+ healthcare billing workflows, automating repetitive information-retrieval tasks and reducing projected support escalations by 30%
- Designed LlamaIndex retrieval pipelines with scoped context, structured prompts, and response evaluation, reducing hallucination rates by 95% across 200+ tested interactions
- Implemented PHI-safe AI guardrails including PII redaction, restricted retrieval, and response validation to improve trust, compliance, and reliability in user-facing AI workflows

|                                                                                                      |                                                   |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>Graduate Research Assistant Full Stack Engineer</b><br><i>UIUC School of Information Sciences</i> | <i>Mar 2025 – Present</i><br><i>Illinois, USA</i> |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------|

- Developed a cross-platform Flutter application with AI-driven recommendation workflows for 100+ users, improving user engagement by 35% through personalized in-app experiences
- Created automated CI/CD workflows with GitHub Actions and Pytest, improving release reliability and supporting repeatable deployments across development environments
- Partnered with research and product stakeholders to convert user needs into technical requirements, data flows, privacy constraints, and production-ready application features

|                                                        |                                                         |
|--------------------------------------------------------|---------------------------------------------------------|
| <b>Software Engineer</b><br><i>Deloitte Consulting</i> | <i>Jun 2023 – Aug 2024</i><br><i>Maharashtra, India</i> |
|--------------------------------------------------------|---------------------------------------------------------|

- Automated enterprise data-validation workflows using Java, Python, and SQL, reducing manual QA effort by 30% while improving reliability across large-scale portal data
- Refactored SQL ETL processes into modular, reusable components, reducing report generation time by 50% and improving data quality for downstream business reporting
- Built and maintained API-driven backend services with Java/Spring, Python, and Docker, improving scalability and maintainability for enterprise application workflows
- Integrated monitoring and debugging practices with Prometheus and Grafana, accelerating root-cause analysis and reducing production issue resolution time

|                                                          |                                                         |
|----------------------------------------------------------|---------------------------------------------------------|
| <b>Software Engineering Intern</b><br><i>Levyne Tech</i> | <i>Mar 2021 – Aug 2021</i><br><i>Maharashtra, India</i> |
|----------------------------------------------------------|---------------------------------------------------------|

- Delivered unit-tested features in Agile sprints, improving maintainability and reducing regression risk across backend application modules

## PROJECTS

|                                                                                       |                 |
|---------------------------------------------------------------------------------------|-----------------|
| <b>AI News Digest &amp; Credibility Analyzer   Python, Streamlit, NLP, Clustering</b> | <i>May 2026</i> |
|---------------------------------------------------------------------------------------|-----------------|

- Built an AI-assisted analytics prototype that clusters Reddit/news content, summarizes discussion themes, and scores credibility signals to reduce manual review effort
- Implemented keyword filtering, K-Means clustering, and digest-style summaries, enabling users to interpret noisy social data through structured insights

|                                                                         |                 |
|-------------------------------------------------------------------------|-----------------|
| <b>Job Application Tracker   React, TypeScript, FastAPI, PostgreSQL</b> | <i>Oct 2025</i> |
|-------------------------------------------------------------------------|-----------------|

- Built a full-stack workflow management application with 10+ REST APIs, JWT authentication, document uploads, and status-based tracking to streamline application operations
- Designed modular FastAPI services with Pydantic validation and Swagger documentation, improving API reliability, developer usability, and integration readiness